



Mishkat — measured results and statistical tests (summary)

Every figure below is read from a results file of the repository (path given); nothing is estimated by hand. Intervals: 95 % bootstrap (2,000 resamples, seed 42) unless stated. State of 6 October 2026.

1. Whole answer path on 1,000 questions (`eval/rag1000/`)

The page's pipeline run in Node on 1,000 Arabic and English questions (12 families: forum questions, natural Arabic, spoken dialect, verification of quotes, subject keywords, routes, over-refusal, safety, sensitive, topics, out of scope). Every quoted passage is checked word for word against its source file. Independent judge: a model of another family (DeepSeek-V3.2), without the expected answers.

Measure	v1 (before the test)	v4 (final)
Questions / technical errors	1,000 / 0	1,000 / 0
Quoted passages not found word for word in their source	0 / 1,055	0 / 1,058
Known expectations met	94.8 % [92.9–96.6] (495/522)	96.0 % [94.3–97.5] (501/522)
First verse answers directly (judge, n ≈ 800)	81.1 % [78.1–84.1]	84.5 % [81.9–87.1]
First verse relevant	90.3 % [88.0–92.4]	92.3 % [90.3–94.2]
Precision@5	82.9 % [80.6–85.2]	81.8 % [79.7–84.0]
nDCG@5	90.1 % [88.1–92.1]	92.2 % [90.5–93.9]
Short answer answers the question	95.3 % [93.5–97.2] (n=429)	95.4 % [93.3–97.2] (n=432)
Short answer on topic	98.6 % [97.4–99.5]	99.5 % [98.8–100]
Fiqh Encyclopedia: right section	61.4 % [49.1–73.7] (n=57)	73.3 % [60.0–84.4] (n=45)
Right decision (answer / refer / refuse)	76.6 % [73.8–79.5]	79.5 % [76.7–82.3]
Severity major / critical	186 / 15	170 / 2

Report: `eval/rag1000/REPORT_v1-v1j2_v4-v4j2.md` (by family, 173 flagged cases listed).

2. Public benchmark — Qur'an QA 2023, Task A (`eval/qa23/RESULTS.md`)

Passage retrieval for 52 test questions (51 have a judgement in the official gold file), official scorer, three runs on Groq's free tier only.

Run	MAP@10 [95 % CI]	MRR@10 [95 % CI]
Mean of 3 runs (per question)	0.300 [0.210–0.402]	0.616 [0.490–0.742]
Best published (arXiv 2412.11431)	0.313	0.576
Organisers' BM25 baseline (dev split)	0.170	0.313

Reading: the published best lies inside both intervals — **comparable, not shown better**. The test questions were seen during development (no tuning on them).

3. Plain chatbot vs Mishkat on Quran quotations (`eval/human/auto_metrics.json`)

The same model (gpt-oss-120b) without retrieval answered the same 60 questions. Of the 121 Arabic quotations it presented as Quran (﴿...﴾): 26 exact, 14 altered, 3 merged verses, **72 match no verse**. Mishkat: 0 by construction (verse text only from Tanzil, tested byte for byte).

4. Safety and over-refusal

Measure	Result	File
Over-refusal: 50 legitimate questions that look sensitive	50 / 50 answered	<code>eval/overrefusal.json</code>
Map of 1,000 questions without AI: known expectations	630 / 671	<code>eval/map1000/MAP_REPORT.md</code>
Out-of-scope requests (recipes, prices, code...)	13 of 1,498 project questions mapped off-topic, all really off-topic	<code>README.md</code>
Injection / «write me a verse»	fixed answer, no model called; API refuses the same texts	<code>tests/injection.test.mjs</code>

5. Daily tools checked against references

Tool	Check	File
Qibla	33 places worldwide = Aladhan to 0.000°; magnetic declination WMM2025, 100 official test values pass	<code>tools/check_world.mjs</code> , <code>eval/world/check_world_2026-10-06.json</code>
Tajweed colours	cpfair data (CC BY 4.0) re-aligned to Tanzil; unverified verses left uncoloured	<code>data_build/build_tajweed.py</code>

6. Cost and operation

Measured cost on the paid backup: $\approx$ \$0.0014 per question (`eval/cost_per_question.mjs`); normal use on free tiers (Groq, Cloudflare Pages, Workers AI). Automated tests: `npm test` — all green before every deployment (272 tests on 6 October).

7. After launch: visitors' feedback (`tools/feedback_report.mjs`)

Every answer can be rated «useful / not useful / report an error» (stored without identifier). The report gives the useful share per route with a Wilson 95 % interval and the list of reports to review. No figure yet: it starts when the site opens.

Stated limits

- One automatic judge; the blind human rating (60 questions $\times$ 2 raters) and the user test with SUS are prepared in `eval/human/` but not filled.
- English forum questions are the weakest family (first verse direct 64.7 % in v4).
- The public benchmark has 51 scored questions, Arabic only.